1.15 Graph Sketching: Cubic Graphs

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Polynomial Graphs

4

In Chapter 2 we briefly saw that a **polynomial** expression is of the form:

$$a + bx + cx^2 + dx^3 + ex^3 + \dots$$

where $a, b, c, d, e, \dots$ are constants (which could be 0).

The **order** of a polynomial is its highest power.

Order	Name
0	
1	
2	
3	
4	
5	

These are covered in Chapter 5.

Chapter 2 explored the graphs for these.

We will cover these now.

While these are technically beyond the A Level syllabus, we will look at how to sketch polynomials in general.

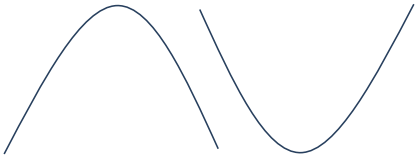
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Polynomial Graphs

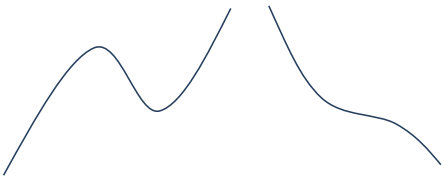
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Order:

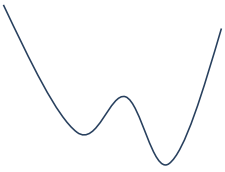
2



3



4



What property connects the order of the polynomial and the shape?

Bro Note: …Actually this is not strictly true, e.g. consider $y = x^4$, which has a U shape. But this is because multiple turns are being squashed into a single point.

In Chapter 2 how did we tell what way up a quadratic is, and why does this work?

1.15 Graph Sketching: Cubic Graphs

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Polynomial Graphs

6

e.g. If $y = 2x^2 + 3$, try a large positive value like $x = 1000$. We can see we'd get a large positive y value. Thus as $x \to \infty$, $y \to \infty$

Equation	If $a > 0$	Resulting Shape	If $a < 0$	Resulting Shape
$y = ax^2 + bx + c$	As $x \to \infty, y \to \infty$ As $x \to -\infty, y \to \infty$		As $x \to \infty, y \to -\infty$ As $x \to -\infty, y \to -\infty$	
$y = ax^3 + bx^2 + cx + d$				
$y = ax^4 + bx^3 + cx^2 + dx + e$				
$y = ax^5 + bx^4 + \ldots$				

If a>0, what therefore can we say about the shape if:
- The order is odd:
- The order is even:

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Cubics

7

Sketch the curve with equation
$y = (x - 2)(1 - x)(1 + x)$

Features you must consider:

Shape?

Roots?

y-intercept?

y

x

Tip: It's incredibly easy to forget to write in one of the intercepts. So don't!

Sketch the curve with equation
$y = x^2(x - 1)$

Shape?

Roots?

y-intercept?

y

x

1.15 Graph Sketching: Cubic Graphs

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Cubics

8

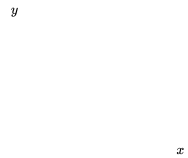
Sketch the curve with equation

$$y = (2 - x)(x + 1)^2$$

Shape?

Roots?

y-intercept?



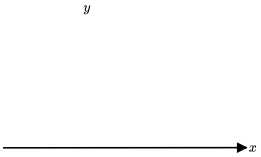
Sketch the curve with equation

$$y = (x - 4)^3$$

Shape?

Roots?

y-intercept?



Exam Notes: The term ‘point of inflection’ has been removed from the new A Level syllabus (bad idea!!).

A point of inflection is where the curve goes from ‘convex’ to ‘concave’ (or vice versa), i.e. curves in one direction before and curves in another direction after. You might have encountered these terms in Physics.

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Cubics with Limited Roots

9

Sketch the curve with equation

$$y = (x + 1)\left(x^2 + x + 1\right)$$

Shape?

Roots?

y-intercept?





However, in Chapter 12, we’ll be able to work **turning points** using ‘differentiation’, and hence conclude that it doesn’t have any!

1.15 Graph Sketching: Cubic Graphs

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Finding the equation yourself

10

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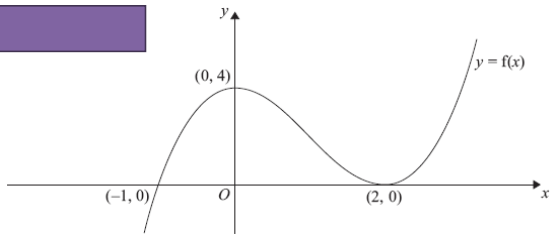


Figure 1 shows a sketch of the curve C with equation $y = f(x)$.
The curve C passes through the point (–1, 0) and touches the x-axis at the point (2, 0).
The curve C has a maximum at the point (0, 4).
The equation of the curve C can be written in the form.

$$y = x^3 + ax^2 + bx + c$$

where a, b and c are integers.
(a) Calculate the values of a, b, c.

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Test Your Understanding

11

1 Sketch the curve with equation $y = x(x-3)^2$

y

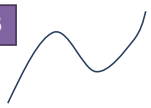
x

2 Sketch the curve with equation $y = -(x+2)^3$

y

x

3



A curve has this shape, touches the x axis at 3 and crosses the x axis at -2. Give a suitable equation for this graph.

Sketch the curve with equation $y = 2x^2(x-1)(x+1)^3$

1.15 Graph Sketching: Cubic Graphs

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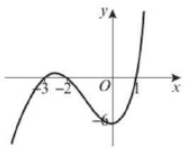
Exercise 4A –Y1 Pure pages 63

12

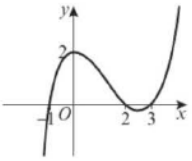
5 The graph of $y = x^3 + bx^2 + cx + d$ is shown opposite, where b, c and d are real constants.

a Find the values of b, c and d.

b Write down the coordinates of the point where the curve crosses the y-axis.



6 The graph of $y = ax^3 + bx^2 + cx + d$ is shown opposite, where a, b, c and d are real constants.
Find the values of a, b, c and d.



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13

7 Given that $f(x) = (x - 10)(x^2 - 2x) + 12x$

a Express $f(x)$ in the form $x(ax^2 + bx + c)$ where a, b, c are real constants.

b Hence factorise $f(x)$ completely.

c Sketch the graph of $y = f(x)$ showing clearly the points where the graph intersects the axes.

1.15 Graph Sketching: Cubic Graphs

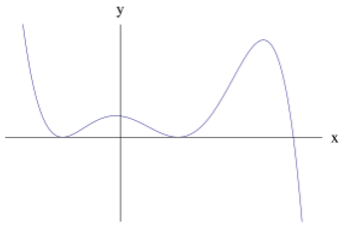
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Exercise 4A Extension

14

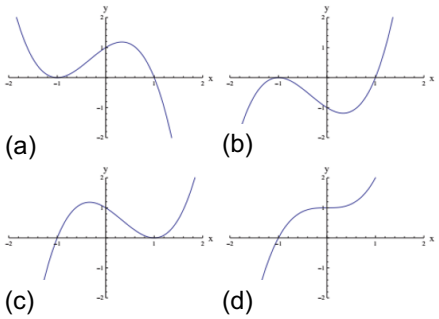
Extension

1 [MAT 2012 1E] Which one of the following equations could possibly have the graph given below?



A) $y = (3-x)^2(3+x)^2(1-x)$
B) $y = -x^2(x-9)(x^2-3)$
C) $y = (x-6)(x-2)^2(x+2)^2$
D) $y = (x^2-1)^2(3-x)$

2 [MAT 2011 1A] A sketch of the graph $y = x^3 - x^2 - x + 1$ appears on which of the following axis?



Cubics can sometimes be factorised by pairing the terms:

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Recap

15

If we sketched $y = (x-a)(x-b)^2(x-c)^3$ what happens on the x-axis at:

$x = a$:

$x = b$:

$x = c$:

a x

b x

c x